

## ROMA CIAMPINO, Italy

WMO: 162390

Lat: 41.8081N

Lon: 12.5847E

Elev: 130

StdP: 99.77

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |             |            | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-0.9 | (c)<br>0.8 | (d)<br>-8.2                   | (e)<br>1.9 | (f)<br>4.6 | (g)<br>-6.0 | (h)<br>2.3 | (i)<br>4.7 | (j)<br>12.0           | (k)<br>10.6 | (l)<br>10.6 | (m)<br>11.2 | (n)<br>2.5            | (o)<br>40 | (p)<br>0.417 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>10.9            | (c)<br>34.0     | (d)<br>21.5 | (e)<br>32.9 | (f)<br>21.5 | (g)<br>31.2 | (h)<br>21.2 | (i)<br>24.7         | (j)<br>28.6 | (k)<br>23.8 | (l)<br>28.3 | (m)<br>23.1 | (n)<br>28.0 | (o)<br>4.1           | (p)<br>210 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.8                            | 18.9 | 26.7 | 22.8 | 17.8 | 26.0 | 21.8 | 16.7 | 25.4 | 75.3          | 28.7 | 71.9 | 28.4 | 68.9 | 28.2 | 28.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| 10.4              | 8.6  | 7.2 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
|                   |      |     | -3.1                       | 36.5 | 1.5                | 1.6 | -4.2                                               | 37.7 | -5.1       | 38.7 | -5.9       | 39.6 | -7.0       | 40.8 |
|                   |      |     | -4.3                       | 26.4 | 1.3                | 1.0 | -5.3                                               | 27.1 | -6.0       | 27.7 | -6.8       | 28.3 | -7.7       | 29.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.1   | 7.8   | 8.5   | 10.9  | 13.9  | 18.0  | 22.5  | 25.3  | 25.5  | 21.2  | 17.4  | 12.8  | 8.9  |
|                                                                                  | DBStd       | 6.80   | 2.90  | 2.83  | 2.52  | 2.51  | 2.60  | 2.93  | 2.15  | 2.24  | 2.55  | 2.62  | 3.19  | 3.04 |
|                                                                                  | HDD10.0     | 224    | 79    | 56    | 20    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 11    | 57   |
|                                                                                  | HDD18.3     | 1517   | 326   | 276   | 231   | 133   | 37    | 2     | 0     | 0     | 5     | 48    | 167   | 291  |
|                                                                                  | CDD10.0     | 2457   | 11    | 13    | 48    | 119   | 249   | 376   | 474   | 481   | 336   | 230   | 95    | 23   |
|                                                                                  | CDD18.3     | 708    | 0     | 0     | 0     | 2     | 28    | 129   | 216   | 223   | 91    | 19    | 1     | 0    |
|                                                                                  | CDH23.3     | 6368   | 0     | 0     | 0     | 13    | 207   | 1156  | 2179  | 2181  | 563   | 69    | 1     | 0    |
|                                                                                  | CDH26.7     | 2452   | 0     | 0     | 0     | 1     | 33    | 403   | 938   | 949   | 124   | 3     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.8    | 2.8   | 3.0   | 3.1   | 3.0   | 2.8   | 2.8   | 2.7   | 2.5   | 2.6   | 2.5   | 3.0   | 2.8  |
| Precipitation                                                                    | PrecAvg     | 800    | 77    | 76    | 67    | 67    | 51    | 34    | 21    | 37    | 72    | 94    | 112   | 96   |
|                                                                                  | PrecMax     | 1167   | 210   | 198   | 141   | 210   | 134   | 127   | 110   | 130   | 164   | 222   | 244   | 259  |
|                                                                                  | PrecMin     | 518    | 3     | 5     | 19    | 16    | 1     | 2     | 0     | 0     | 4     | 3     | 7     | 27   |
|                                                                                  | PrecStd     | 172    | 55    | 51    | 36    | 50    | 30    | 30    | 27    | 36    | 49    | 69    | 61    | 54   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.4  | 17.2  | 21.1  | 25.0  | 29.9  | 34.0  | 35.2  | 36.2  | 31.8  | 26.9  | 22.1  | 17.9 |
|                                                                                  |             | MCWB   | 12.2  | 11.2  | 12.7  | 16.1  | 18.8  | 21.5  | 21.9  | 21.7  | 20.4  | 18.9  | 16.9  | 14.4 |
|                                                                                  | 2%          | DB     | 15.0  | 15.5  | 18.9  | 22.8  | 27.1  | 32.3  | 33.9  | 34.8  | 29.5  | 25.1  | 20.2  | 16.0 |
|                                                                                  |             | MCWB   | 12.4  | 11.0  | 12.9  | 15.0  | 17.9  | 21.3  | 21.6  | 21.7  | 20.7  | 18.8  | 16.5  | 13.7 |
|                                                                                  | 5%          | DB     | 13.9  | 14.2  | 17.2  | 20.9  | 25.8  | 30.9  | 32.8  | 33.0  | 28.0  | 23.9  | 19.1  | 15.0 |
|                                                                                  |             | MCWB   | 11.5  | 10.5  | 12.0  | 14.4  | 17.5  | 20.8  | 21.4  | 21.9  | 20.0  | 18.6  | 15.9  | 12.8 |
|                                                                                  | 10%         | DB     | 12.8  | 13.2  | 16.0  | 19.2  | 24.0  | 29.1  | 31.2  | 31.2  | 26.6  | 22.2  | 18.0  | 14.0 |
|                                                                                  |             | MCWB   | 10.5  | 10.0  | 11.6  | 13.7  | 17.2  | 20.1  | 21.1  | 21.4  | 19.8  | 18.0  | 15.2  | 11.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.0  | 13.1  | 15.4  | 17.7  | 21.0  | 24.9  | 25.7  | 26.0  | 24.0  | 21.9  | 18.7  | 15.4 |
|                                                                                  |             | MCDB   | 15.1  | 15.1  | 18.2  | 22.0  | 25.8  | 29.0  | 29.6  | 29.8  | 27.0  | 24.2  | 20.2  | 16.8 |
|                                                                                  | 2%          | WB     | 12.8  | 12.2  | 14.0  | 16.3  | 19.7  | 23.4  | 24.5  | 24.9  | 22.8  | 20.6  | 17.7  | 14.1 |
|                                                                                  |             | MCDB   | 14.4  | 14.2  | 16.8  | 20.5  | 24.5  | 28.5  | 28.8  | 28.7  | 26.1  | 22.9  | 19.3  | 15.6 |
|                                                                                  | 5%          | WB     | 11.8  | 11.6  | 13.1  | 15.3  | 18.8  | 22.3  | 23.7  | 24.0  | 21.9  | 19.6  | 16.6  | 13.2 |
|                                                                                  |             | MCDB   | 13.5  | 13.6  | 15.8  | 19.3  | 23.4  | 27.7  | 28.5  | 28.2  | 25.4  | 22.1  | 18.5  | 14.7 |
|                                                                                  | 10%         | WB     | 10.9  | 10.7  | 12.3  | 14.5  | 17.9  | 21.3  | 22.9  | 23.2  | 21.1  | 18.7  | 15.5  | 12.1 |
|                                                                                  |             | MCDB   | 12.5  | 12.8  | 15.1  | 18.2  | 22.4  | 26.8  | 28.1  | 28.0  | 24.8  | 21.3  | 17.5  | 13.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.8   | 8.5   | 9.1   | 9.3   | 9.9   | 10.6  | 11.0  | 10.9  | 9.8   | 9.1   | 7.7   | 7.5  |
|                                                                                  |             | MCDBR  | 7.7   | 9.3   | 10.9  | 11.6  | 12.3  | 12.3  | 12.6  | 13.0  | 11.2  | 10.3  | 8.0   | 7.0  |
|                                                                                  |             | MCWBR  | 5.6   | 6.1   | 6.5   | 6.2   | 5.6   | 5.4   | 5.7   | 5.6   | 5.3   | 5.9   | 5.5   | 5.2  |
|                                                                                  | 5% WB       | MCDBR  | 6.9   | 7.5   | 9.2   | 9.8   | 10.9  | 11.1  | 10.8  | 10.8  | 9.4   | 8.7   | 6.7   | 6.2  |
|                                                                                  |             | MCWBR  | 5.6   | 5.7   | 6.2   | 6.2   | 6.1   | 5.8   | 6.0   | 6.0   | 5.7   | 6.0   | 5.4   | 5.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.350  | 0.376 | 0.418 | 0.443 | 0.448 | 0.440 | 0.431 | 0.431 | 0.423 | 0.412 | 0.386 | 0.351 |      |
|                                                                                  | taud        | 2.399  | 2.301 | 2.180 | 2.134 | 2.169 | 2.244 | 2.274 | 2.285 | 2.299 | 2.292 | 2.295 | 2.390 |      |
|                                                                                  | Ebn at Noon | 784    | 817   | 821   | 828   | 833   | 838   | 842   | 830   | 808   | 765   | 725   | 746   |      |
|                                                                                  | Edh at Noon | 82     | 104   | 132   | 148   | 147   | 137   | 131   | 126   | 116   | 105   | 90    | 77    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.75   | 2.66  | 3.90  | 5.19  | 6.35  | 7.23  | 7.41  | 6.46  | 4.71  | 3.22  | 1.98  | 1.58  |      |
|                                                                                  | RadStd      | 0.17   | 0.27  | 0.45  | 0.36  | 0.53  | 0.31  | 0.34  | 0.35  | 0.23  | 0.28  | 0.18  | 0.17  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | +0.45 | N/A     | N/A    | N/A     | -0.50 | -0.83 | N/A         | -77     | N/A     | N/A     |
| (47) Regional (18 neighbors) |  | +0.42 | N/A     | N/A    | +0.53   | N/A   | N/A   | N/A         | -78     | +135    | +69     |

Nomenclature: See separate page